

PHYS 357: Honours Quantum Physics 1

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August 31st, 2026

Abstract

NOT FINISHED

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1 Introduction

Textbook used is JS Townsend A Modern Approach to Quantum Mechanics, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on wednesdays, best 8 of 12).
- 20% midterms (best of 2, october 2nd and november 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 Angular Momentum and Magnetic Moments

2.1.1 Angular Momentum

Definition 1 (Orbital Angular Momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (1)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr. \quad (2)$$

Definition 2 (Angular Momentum of a Rigid Body). *For a rigid body rotating about a fixed axis this*

$$L = I_0\omega, \quad (3)$$

with I_0 the moment of inertia and ω the angular velocity.

2.1.2 Magnetic moment of a circulating charge

Definition 3 (Magnetic moment). *A magnetic moment is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

Consider a point particle of mass m and charge q in a circular orbit of radius r and period T . Three steps connect this to a magnetic moment.

1. The charge passes any point on the orbit once per period, so it is equivalent to a steady current

$$I = \frac{q}{T} = \frac{qv}{2\pi r}.$$

2. A current loop of area $A = \pi r^2$ has magnetic moment

$$\mu = \frac{IA}{c}.$$
¹

3. Substituting gives

$$\mu = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z. \quad (4)$$

Result 1 (Gyromagnetic Ratio). *The magnetic moment is proportional to the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \quad \gamma = \frac{q}{2mc}. \quad (5)$$

The constant γ is the gyromagnetic ratio.

Remark 1. *This is a purely classical result. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

September 2,
2026

2.2 Spin and the Magnetic Moment

Remark 2. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

In quantum mechanics the intrinsic angular momentum $\vec{S}$, called spin, replaces the orbital angular momentum $\vec{L}$. The magnetic moment is still proportional to it, but with an extra dimensionless factor,

$$\vec{\mu} = \frac{qg}{2mc} \vec{S}, \quad (6)$$

where g is the g-factor of the particle. It has no classical analogue and must be measured or derived from relativistic quantum theory.

Definition 4 (Gyromagnetic ratio). *The gyromagnetic ratio is the ratio of the magnetic moment to the angular momentum,*

$$\gamma = \frac{qg}{2mc}, \quad (7)$$

where q is the charge, g is the g-factor, m is the mass, and c is the speed of light.

Definition 5 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (8)$$

It carries units of angular momentum, so $[\hbar] = [L_z]$.

¹Here c is in Gaussian units.

2.2.1 Magnetons

Factoring $\hbar$ out of the gyromagnetic ratio gives a natural unit of magnetic moment for each mass scale.

Definition 6 (Bohr magneton).

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (9)$$

Definition 7 (Nuclear magneton).

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (10)$$

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (11)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

2.2.2 Values for the three particles

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 3. *The neutron is neutral but still has a magnetic moment, so $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.*

Remark 4. *Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.*

The force is just $\vec{F} = -\vec{\nabla}U$, $F_z = -\frac{\partial U}{\partial z}$. $U = -\vec{\mu} \cdot \vec{B}$. For Silver (Ag) $\vec{L} = 0$, so $\vec{S} \neq 0 \rightarrow$ single electron, $U = -g_e \frac{|e|\hbar}{2m_e c} S_z B_z$. $F_z = -\frac{g_e |e|\hbar}{2m_e c} \frac{dB_z}{dz} S_z$, $F_z = \frac{g_e |e|\hbar}{2m_e c} \frac{dB_z}{dz} |S_z|$, $F_z S_z$.

A Appendix

B Useful Links

Stern-Gerlach experiment: <https://www.youtube.com/shorts/QG7kJXxN1Kk>